

Birla Institute of Technology and Science, Pilani

Work Integrated Learning Programmes Division

Cluster Programme - M.Tech. in Artificial Intelligence and Machine Learning.

II Semester 2025-26

Course Number	AIML ZC416	
Course Name	Mathematical Foundations for Machine Learning	
Nature of Exam	Open Book	# Pages 3
Weightage for nablaing	40%	# Questions 5
Duration	150 minutes	
Date of Exam	24/05/2026 (FN)	

Instructions

1. All questions are compulsory
 2. Write the question number clearly before answering a question.
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Q1. Answer the following questions with justifications.

- (1) Unrolling the update step for momentum-based gradient descent: $x_{i+1} = x_i - \eta[\nabla f(x_i)]^T + \beta(x_i - x_{i-1})$, obtain an expression for x_{i+1} using only gradients computed from the 0th step to the the current step. [3 Marks]
- (2) We would like to perform PCA on a bunch of points in a plane as follows: $z = (x, y)^T = (1, 1)^T + u(2, 0)^T + v(0, 1)^T$ where u and v are Gaussian random variables with mean 0 and variance 1. Calculate the data covariance matrix of the mean-centred data. You may use $E(u^2) = 1$ and $E(v^2) = 1$ in your calculations. [4 Marks]

Q2. Answer the following questions with justifications.

- (1) For the problem below, explain why strong duality holds. Note that a and b are positive constants: [1 Mark]

$$\min x + y \text{ subject to } ax^2 + by^2 \leq 1$$

- (2) Set up the Karash-Kuhn Tucker (KKT) conditions for the optimization problem in part (1) of this question. [2 Marks]
- (3) Use the KKT conditions to find the optimal solution to the optimization problem in part (1) of this question. [4 Marks]

Q3. Answer the following questions with justifications.

- (1) Consider a binary classification problem in $\mathbb{R}^2$ where the two possible classes are +1 and -1. The SVM linear classifier is of the form $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$, where $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $b \in \mathbb{R}$.
The following three support vectors (along with their true class) that arise out of SVM classification problem are:

- i) $\mathbf{x}_A = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $y_A = +1$
 ii) $\mathbf{x}_B = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $y_B = -1$
 iii) $\mathbf{x}_C = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$, $y_C = -1$

Calculate the values of w_1, w_2 and b by using the support vectors given above. [5 marks]

- (2) Consider a binary classification problem where the two possible classes are +1 and -1. You are given a linear classifier defined as $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$ where $\mathbf{w} = [1 \ 1]^T$ and $b = -3$. Calculate hinge loss for the binary classification problem (where the prediction is given by $f(\mathbf{x})$) for the following four data points: $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ and $\mathbf{x}_4$.

- (a) $\mathbf{x}_1 = [3 \ \frac{1}{4}]^T$
 (b) $\mathbf{x}_2 = [3 \ 2]^T$
 (c) $\mathbf{x}_3 = [0 \ 0]^T$
 (d) $\mathbf{x}_4 = [1 \ \frac{1}{2}]^T$

The true class of $\mathbf{x}_1, \mathbf{x}_2$ is '+1' and of $\mathbf{x}_3, \mathbf{x}_4$ is '-1'. [2 marks]

Q4. Consider the following dataset associated with a binary classification problem in $\mathbb{R}^2$ where the two possible classes are +1 and -1. [7 Marks]

Input	True Class
$\begin{bmatrix} -1 \\ -1 \end{bmatrix}$	+1
$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$	-1
$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$	-1
$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	+1

Observe that this dataset is NOT linearly separable. Derive a Kernel-SVM based classifier for this problem by using the kernel $K(\mathbf{x}, \mathbf{z}) = (1 + x_1 z_1 + x_2 z_2)^2$ where $\mathbf{x} = [x_1 \ x_2]^T$ and $\mathbf{z} = [z_1 \ z_2]^T$. Note that this kernel function corresponds to the feature map $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^6$ defined as:

$$\phi\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = [1 \ x_1^2 \ \sqrt{2}x_1x_2 \ x_2^2 \ \sqrt{2}x_1 \ \sqrt{2}x_2]^T$$

Q5. Answer the following questions with justifications.

(1) Is it true that the algebraic multiplicity of an eigenvalue is always equal to its geometric multiplicity? [1 Mark]

(2) Evaluate the gradient of $\mathbf{x}^T \mathbf{A} \mathbf{x}$ with respect to $\mathbf{x}$. [1 Mark]

(3) Is it true that for all matrices $\mathbf{A}$, the matrix $\mathbf{A}^T \mathbf{A}$ is positive definite? Note that $\mathbf{A}$ need not be a square matrix. [1 Mark]

(4) If matrices $\mathbf{A}$ and $\mathbf{B}$ are constructed such that $\mathbf{AB} = \mathbf{I}_n$, where $\mathbf{I}_n$ is a identity matrix of order n . Do we conclude that $\mathbf{B}$ is the inverse of $\mathbf{A}$ or vice-versa? [1 Mark]

(5) Find, if possible, vectors $\mathbf{x}, \mathbf{y}$ and an inner product $\langle, \rangle$ such that

$$\langle \mathbf{x}, \mathbf{y} \rangle > \|\mathbf{x}\| \cdot \|\mathbf{y}\|$$

, where $\|\cdot\|$ is the norm induced by the inner product. [1 Mark]

(6) Find, if possible, the decomposition of the given square matrix $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ as sum of symmetric and skew-symmetric matrices. [1 Mark]

(7) Why a linear system of equation $\mathbf{Ax} = \mathbf{b}$ cannot have exactly three solutions? [1 Mark]

(8) Let $\mathbf{V} = \mathbb{R}^2$ be a vector space over $\mathbb{R}$. Find if $\mathbf{W} = \{(x, 2) | x \in \mathbb{R}\}$ is a subspace of $\mathbf{V}$. [1 Mark]

(9) Given a fast algorithm that generates the eigenvalues and corresponding eigenvectors of a matrix, construct, if possible, an algorithm to get matrices that are orthogonal. [2 Marks]

(10) Solve the following system of equations [2 Marks]

$$\begin{bmatrix} 1 & 0 & 8 & -4 \\ 0 & 1 & 12 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 42 \\ 8 \end{bmatrix}$$